

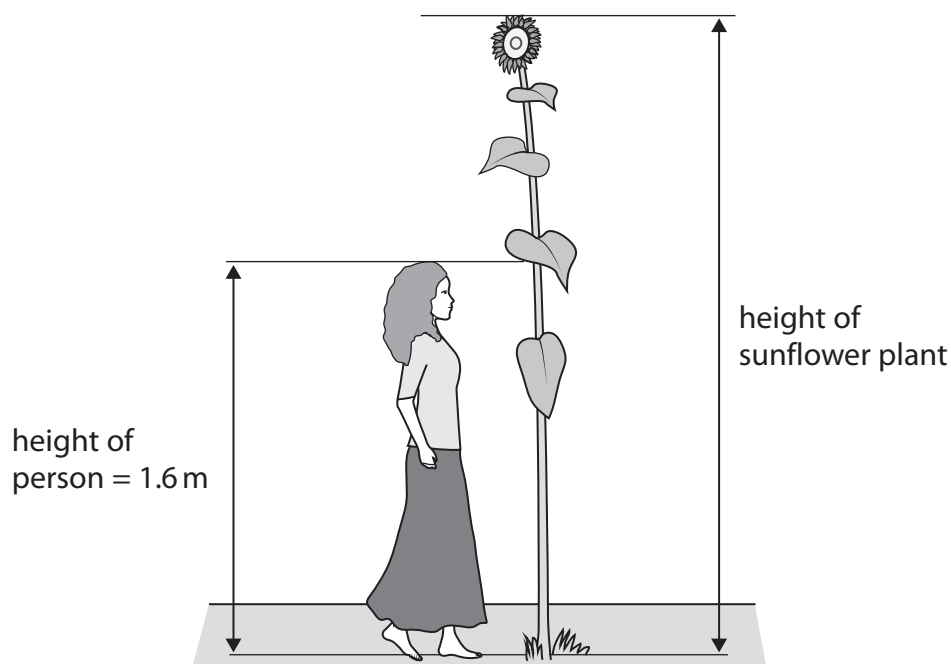
Answer ALL questions.

Write your answers in the space provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1** Plants contain tissues involved in the transport of substances.

The diagram shows a sunflower plant.



- (a) Estimate the height of the sunflower, using the information in the diagram.

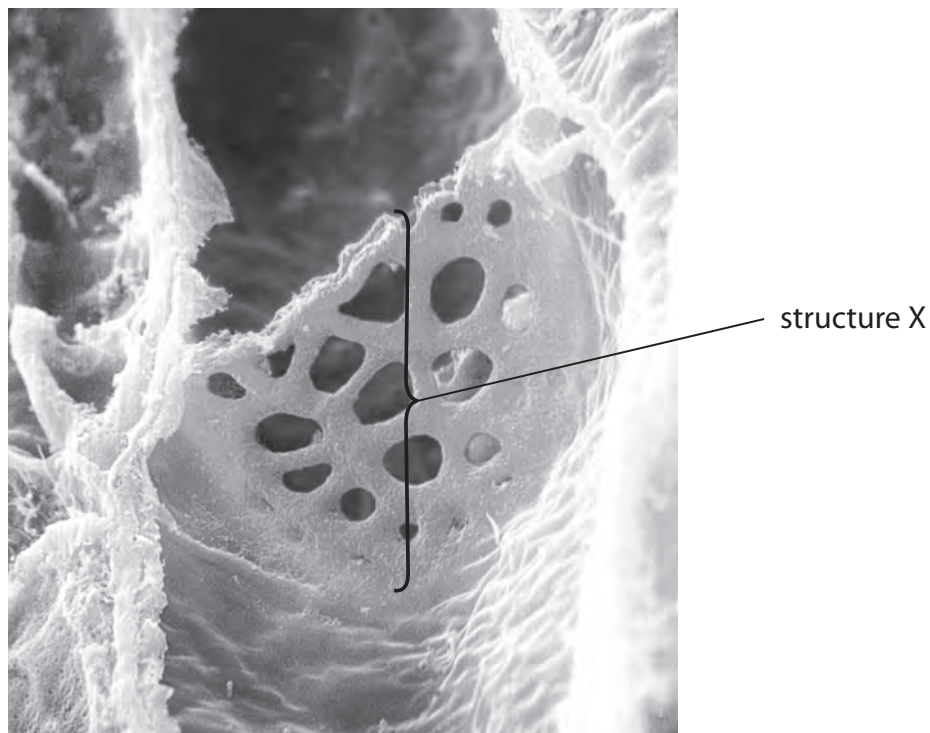
Give your answer with appropriate units.

(1)

Answer



(b) The photograph shows a structure that would be found in a sunflower plant.



(Source: © Dr. Richard Kessel and Dr. Gene Shih / Science Photo Library)

(i) Name the tissue that would contain structure X.

(1)

(ii) Describe how **two** structures in this tissue enable it to carry out its function.

(2)

Structure 1

.....

.....

Structure 2

.....

.....



(c) The photograph shows some rings of lignin, as seen using a microscope.



(Source: © Science Photo Library / Alamy Stock Photo)

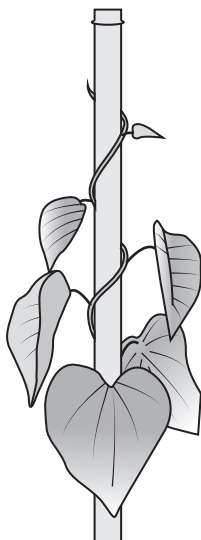
(i) Which of the following tissues contain lignin?

1. phloem
2. sclerenchyma
3. xylem

(1)

- ☐ **A** 1 only
- ☐ **B** 1 and 3 only
- ☐ **C** 2 and 3 only
- ☐ **D** 3 only

(ii) The diagram shows a climbing plant that is growing up a bamboo pole.



Suggest **two** reasons why a climbing plant may have a **lower** lignin content than a sunflower plant of the same height and diameter.

(2)

(Total for Question 1 = 7 marks)